

Discovery and Alignment First, Then Orchestration and Execution

INTRODUCTION

Orchestration is a key concept in modern computing. Workflow engines, cloud platforms, service ecosystems, and AI agents rely on orchestration to coordinate computational capabilities so that computations can be executed to meet desired goals. Historically, orchestration systems have largely assumed that the capabilities to be coordinated are already known. As machines gain access to increasingly large universes of computable knowledge, this assumption becomes less tenable. Before orchestration can occur, machines first need information about what computable knowledge is available. In general terms, web-scale orchestration requires pre-orchestration mechanisms that identify and characterize computational possibilities prior to orchestration and execution.

ORCHESTRATION AND ITS ASSUMPTIONS

Traditional orchestration systems answer a straightforward question:

Given a collection of computational capabilities, how should they be coordinated?

Examples include workflow engines, service orchestration platforms, fixed data analytic pipelines, container orchestration systems, scientific workflow managers, and modern agent frameworks. Although these systems differ substantially in implementation, they generally share an important assumption: the capabilities to be orchestrated are already known or can be easily identified.

This assumption has held up for decades. Workflows, services, software packages, APIs, analytic pipelines and computational resources generally exist in manageable systems. Existing orchestration typically operates over computational capability sets that are relatively constrained. As the number and diversity of available computational capabilities increase, determining which capabilities may contribute to a specified goal becomes an increasingly important computational problem in its own right. Thus the question shifts from "How should known capabilities be coordinated?" to "Which capabilities are relevant enough to coordinate in the first place?"

The emergence of generative AI systems is expanding the role of orchestration in important ways. Systems based on large language models increasingly accept goals expressed in natural language rather than explicit workflows as inputs. Tool selection, workflow construction, and execution planning may be performed dynamically at runtime. Recent work involving agent frameworks, tool use, model context protocols, AI planning, service composition, and intent-driven systems demonstrates growing interest in machine-mediated orchestration of computational capabilities.

Yet these developments have mostly left a related and fundamental question unaddressed.

THE SCALING CHALLENGE

What happens when the universe of available computable knowledge and its corresponding computational capabilities becomes extremely large?

Consider an environment containing millions of predictive models, classifiers, calculators, recommendation functions, software services, knowledge graphs, and other forms of computable knowledge. In such environments, orchestration encounters a problem that orchestration alone cannot solve.

An orchestrator cannot coordinate capabilities that have not yet been identified.

It cannot arrange capabilities that have not yet been recognized as potentially useful.

It cannot compute with capabilities whose relevance to a goal remains unknown.

Web-scale orchestration requires pre-orchestration so that machines can determine what are actionable computational possibilities.

PRE-ORCHESTRATION ACTIVITIES

The role of pre-orchestration activities is to characterize ever-changing computational capabilities on demand. They are made necessary by changing assumptions about the size and nature of the available universe of computable knowledge and its corresponding computational capabilities.

At a minimum, two activities appear necessary.

Discovery seeks to identify computational capabilities that exist within a very large computable knowledge environment.

Alignment seeks to determine which capabilities may be useful toward realization of a specified goal. Importantly, alignment need not produce a single answer. A goal may align with multiple capabilities, multiple capability combinations, or multiple candidate goal realization pathways. The result may be a list of numerous computational paths rather than a single path.

From this perspective, the primary output of discovery and alignment is not computation that realizes some goal. Rather, the output is information sufficient to determine which computations are possible and potentially relevant to a specified goal. Together, discovery and alignment help determine the scope of current computational capabilities in any large universe of them. They characterize sets of possible computational capabilities before any commitment to a particular method of orchestrating and executing a computation has been made.

Orchestration remains essential, but it operates on capabilities that have already been identified. Orchestration narrows possibility into commitment. Execution then applies computational resources to perform the selected computation.

The distinctions made here are not intended to be rigid ones. Boundaries between discovery, alignment, orchestration, and execution will often blur in practice. Modern foundation-model-based systems already combine aspects of all four activities within single information systems.

Nevertheless, these distinctions highlight an increasingly important reality: before machines can produce wanted computations effectively by taking advantage of large computable knowledge environments, they first need to determine what computations are possible and relevant to realize stated goals.

IMPLICATIONS

The historical focus of orchestration research has been the coordination of known computational capabilities. As computational capability ecosystems continue to expand, an equally important challenge emerges upstream of orchestration itself.

Machines may increasingly require mechanisms for discovering computational capabilities and aligning those capabilities to goals before orchestration and execution can take place.

In this way, the future of machine-mediated computation may depend as much on machine-supported pre-orchestration activities as on orchestration and execution of computations.

The challenge shifts from determining how computation should be performed to determining which computational possibilities are available and relevant to a specified goal.

In this view, orchestration continues to do what it has always done. What changes is that, in web-scale computable knowledge environments, machines increasingly require upstream mechanisms to determine what should be orchestrated.